Machine Learning Fundamentals

The ML Pipeline

A typical ML project follows these steps: (1) define the problem, (2) collect and explore data (EDA), (3) preprocess and feature-engineer, (4) select and train a model, (5) evaluate, (6) tune hyperparameters, (7) deploy, (8) monitor and retrain.

Supervised Learning

In supervised learning, each training example is paired with a label. Regression predicts continuous values (house prices); classification predicts discrete classes (spam vs ham).

Key algorithms: Linear Regression, Logistic Regression, Decision Trees, Random Forest, Gradient Boosted Trees (XGBoost, LightGBM), Support Vector Machines, Neural Networks.

Unsupervised Learning

Unsupervised learning discovers hidden structure in unlabelled data. Clustering groups similar points (K-Means, DBSCAN, Hierarchical Clustering). Dimensionality reduction compresses features (PCA, t-SNE, UMAP, Autoencoders).

Association rule learning finds co-occurrence patterns in transactional data (Apriori, FP-Growth).

Model Evaluation

Regression metrics: MAE, MSE, RMSE, R^2 score.

Classification metrics: accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix.

Cross-validation (k-fold) provides a robust estimate of generalisation performance and reduces variance from a single train/test split.

Overfitting and Regularisation

Overfitting occurs when a model memorises training data but performs poorly on unseen data. Underfitting happens when a model is too simple to capture the underlying pattern.

Regularisation techniques: L1 (Lasso), L2 (Ridge), Elastic Net, dropout (neural nets), early stopping, data augmentation, and increasing training data size.

Feature Engineering

Feature engineering transforms raw data into informative inputs. Techniques include one-hot encoding (categorical variables), scaling (StandardScaler, MinMaxScaler), polynomial features, interaction terms, and target encoding.

Feature selection methods: filter methods (correlation, chi-squared), wrapper methods (RFE), embedded methods (LASSO coefficients, tree feature importance).